

Personality Traits and Emotion Recognition to Predict Engineering Students’ Soft Skills using applications of LLMs

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Background and aims

Soft skills matter in engineering education. This project investigates whether interview language and vocal delivery can support their estimation, using personality as an intermediate step.

Research question

Can LLM-derived text cues and audio characteristics predict interview ratings and support an exploratory Big Five-to-BESSI skill mapping?

Objectives

1. Extract structured language and audio features.
2. Compare predictions with simple baselines and published RecruitView results.
3. Evaluate a separate personality-to-skill mapping and identify its limits.

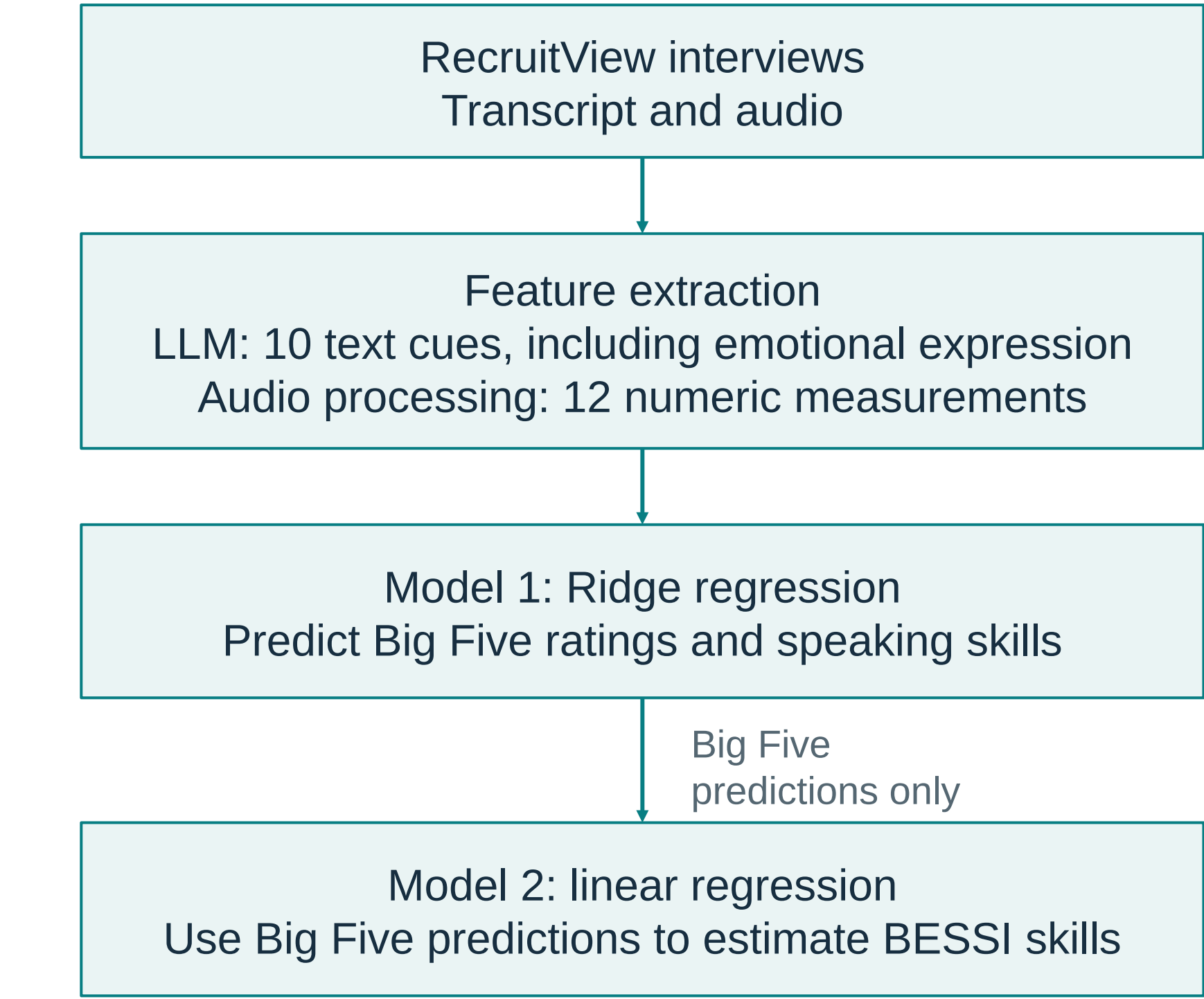
Research basis

RecruitView [1] provides interview personality and performance ratings. BESSI research [2, 3] studies related personality and skill measures. Relatedness supports testing a mapping, not assuming that personality and skills are equivalent.

Data

RecruitView: 1,998 responses, 331 people.
Paired Big Five/BESSI data: 313 usable rows.
No visual features in the final model.

Method and evaluation



Model 2 learns from separate paired questionnaire data. Statistical scale adjustment connects the two models. Speaking skills remains a separate Model 1 output.

Why this approach?

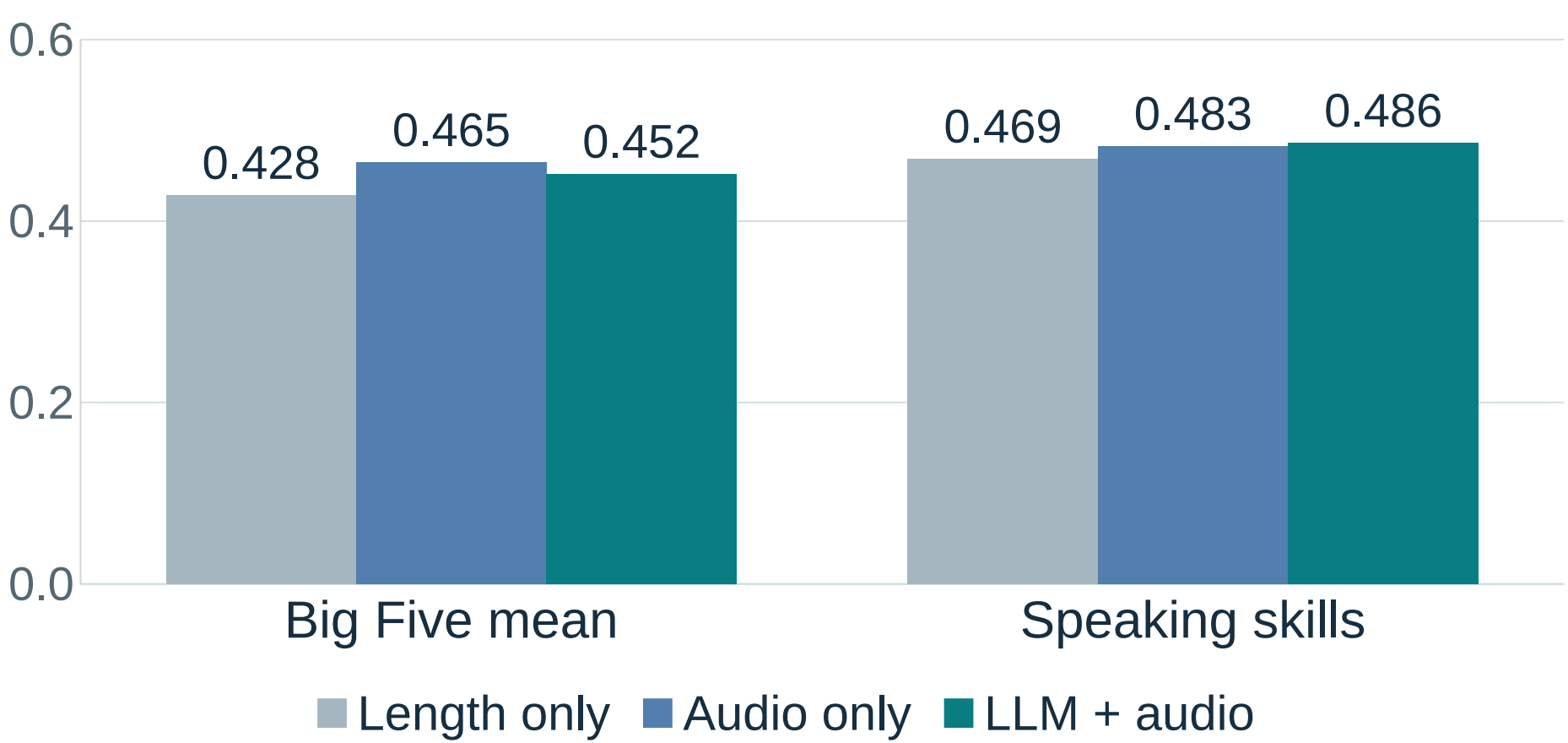
LLMs turn wording into reusable numeric features. Compact regression is interpretable and limits complexity with a small sample.

Evaluation checks

Keep each RecruitView participant in one split.
Choose the model on validation data. Check stability, response length and uncertainty.

Results and interpretation

Interview prediction: higher correlation is better



Spearman correlation, not percentage accuracy.
Test: 292 responses from 49 participants.

Published paper comparison [1]

CRMF: Big Five 0.523; speaking skills 0.595.
The paper also uses video and a different split. This is a contextual comparison, not a reproduction.

Separate BESSI mapping

Mean R² = 0.504 on 47 test rows. Participant independence is unverified. This is not the accuracy of the full interview-to-skills pipeline.

Main finding

The compact model predicts useful patterns. Extra benefit over response length remains uncertain: 95% intervals for the improvements include zero.

Progress and completion plan

Completed: data, models, comparisons and final checks.
Poster deadline: 16 September 2026. Proposed plan at right.
Dissertation writing and review remain; dates to confirm.

Poster completion	11-12 Sep	13-14 Sep	15-16 Sep
Prepare content and layout	Planned		
Seek feedback and revise		Planned	
Final checks and submit PDF			Planned

Limits: no direct BESSI labels in RecruitView; emotion cues lack independent validation; engineering-student generalisation is untested.
Follow-up test analyses are exploratory. Research use only; do not use these estimates for hiring decisions.

[1] Gupta et al. (2025), RecruitView, arXiv:2512.00450. [2] Soto et al. (2022), BESSI, doi:10.1037/pspp0000401.
[3] Sewell et al. (2022), Survey data of social, emotional, and behavioral skills, doi:10.1016/j.dib.2022.107792.